



**Faculty of Engineering & Technology Electrical & Computer Engineering
Department**

ENEE4113 - COMMUNICATION LAB

Experiment #2 PRELAB

Double-side and Single-side Band Modulation

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Section: 4

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- **Theoretical Part:**

Amplitude Modulation (AM): is a technique where the amplitude of a high-frequency carrier signal is varied in proportion to the information signal. The demodulation process involves extracting the original information signal from the modulated signal using envelope detection.

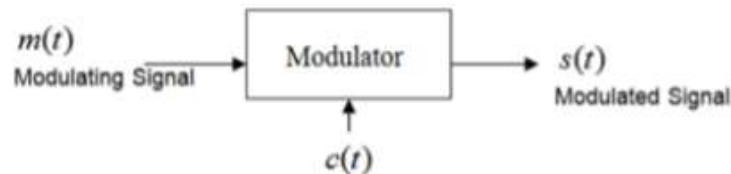


Figure 1 : Modulation Block System

- $S_{AM}(t) = A_c [1 + \mu \cos(\omega_m t)] \cos(\omega_c t)$
- Demodulated signal = $A_c \mu \cos(\omega_m t)$

Double-sideband Suppressed Carrier (DSB-SC) Modulation is a type of AM that eliminates the carrier signal and transmits only the two sidebands. Demodulation involves multiplying the modulated signal with a local oscillator signal of the same frequency and phase as the original carrier signal.

$$s(t) = m(t)c(t) \Rightarrow s(t) = A_m A_c \cos(2\pi f_m t) \cos(2\pi f_c t)$$

Single Sideband (SSB) Modulation is a type of AM that transmits only one of the two sidebands and eliminates the carrier signal. Demodulation involves using a product detector or synchronous detector to recover the original message signal.

$$s(t) = \frac{A_m A_c}{2} \cos [2\pi(f_c + f_m) t] + \frac{A_m A_c}{2} \cos [2\pi(f_c - f_m) t]$$

- **Software Part:**

- **Coding Simulation (Matlab script):**

- **DSBsc :**

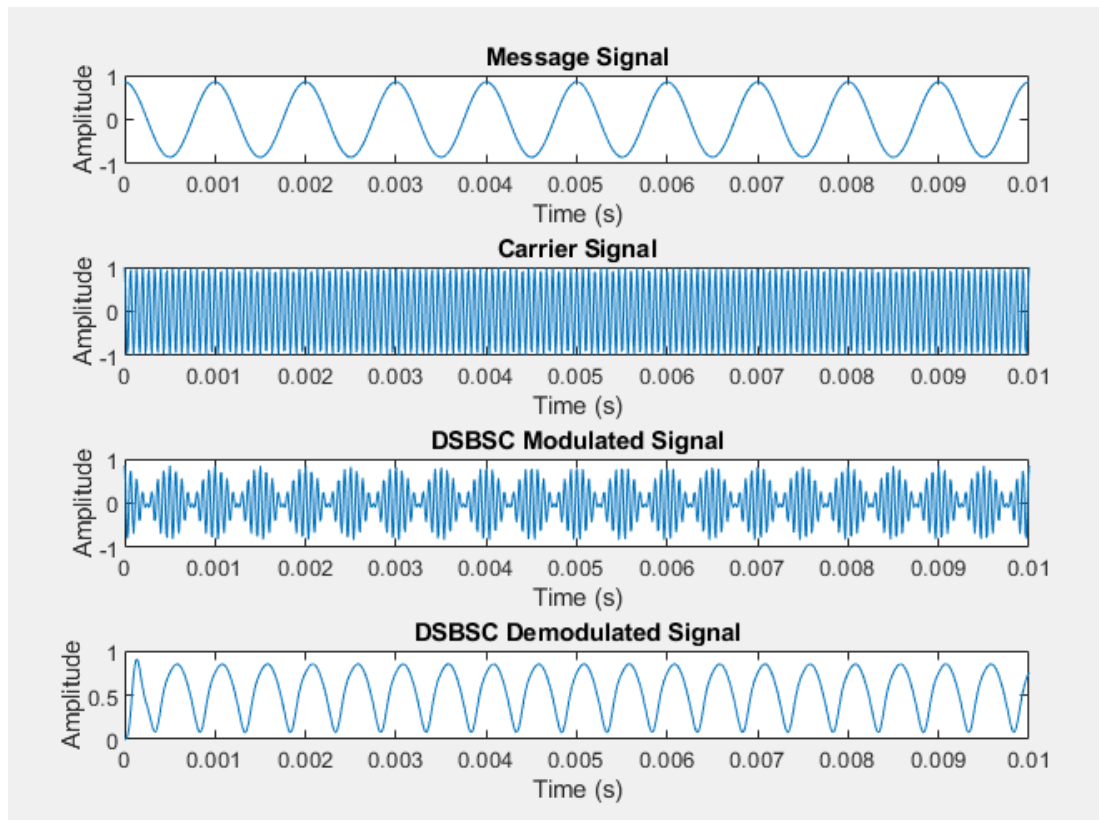


Figure 2 : DSBsc Simulation output

As shown in the above output, the MATLAB code demonstrates DSB-SC AM modulation and demodulation of a message signal with a carrier signal. The modulated signal contains high-frequency components that are removed during demodulation to extract the original message signal. The code creates graphs that show the amplitude and frequency spectrum of each signal in the time and frequency domains. The demodulated signal resembles the original message signal, but with added noise and high-frequency components. The code provides a visual representation of how DSB-SC AM modulation and demodulation work and how they affect the original message signal.

- SSBsc :

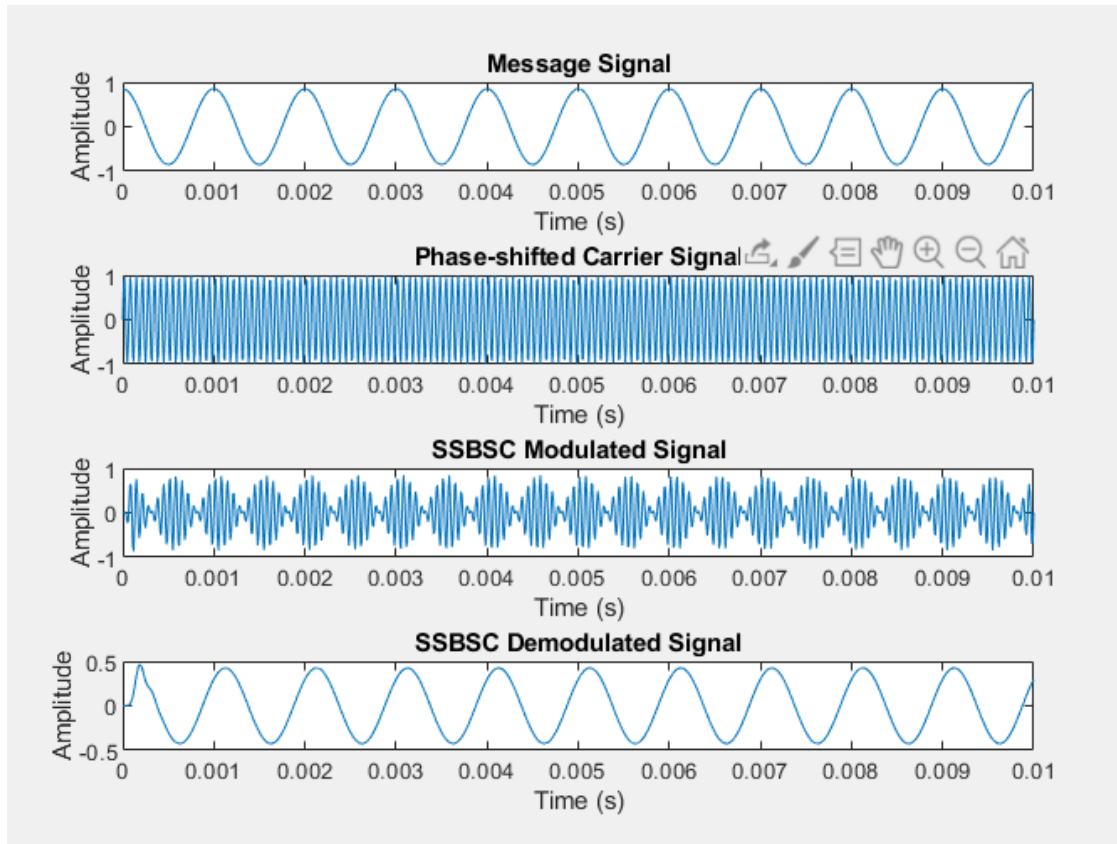


Figure 3 : SSBsc Simulation output

The SSBSC AM modulation and demodulation code produces a clear demonstration of the modulation process where one of the sidebands along with the carrier is removed from the modulated signal. The demodulation process regenerates the original message signal with some noise and distortion. The generated graphs of the message signal, carrier signal, modulated signal, and demodulated signal in both time and frequency domains provide a clear understanding of the SSBSC AM modulation and demodulation process. The demodulated signal is almost identical to the original message signal, but with some noise and distortion, which can be observed from the generated plots.

- **Block Simulation (Matlab Simulink):**

- **DSBsc :**

➤ The block simulation :

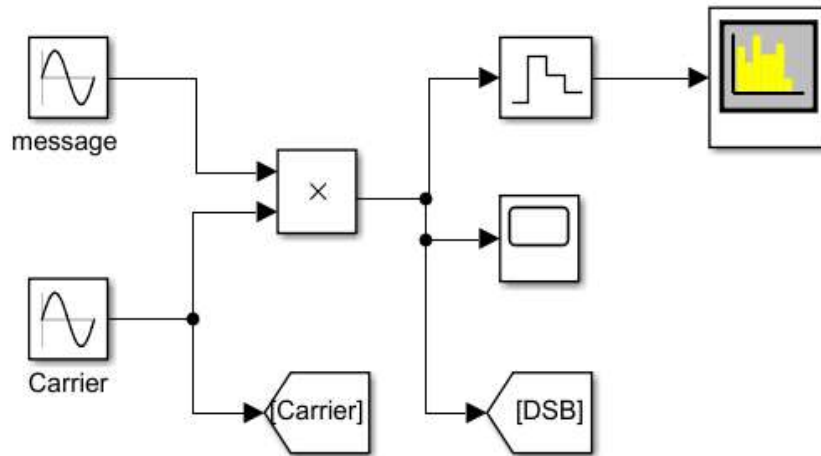


Figure 4: DSBsc modulation block simulation

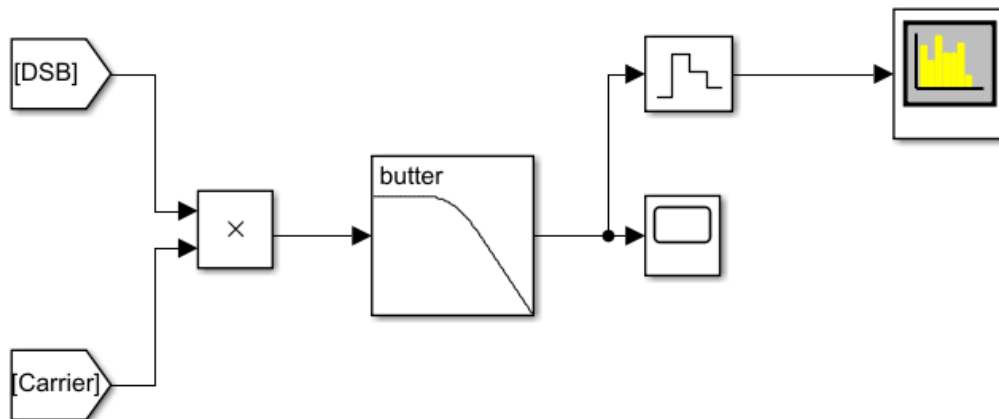


Figure 5: DSB deModulation block simulation

After linking the images above and adjusting the signal, filter and measurement settings. I got these results:

1. The modulation :

➤ In frequency domain:

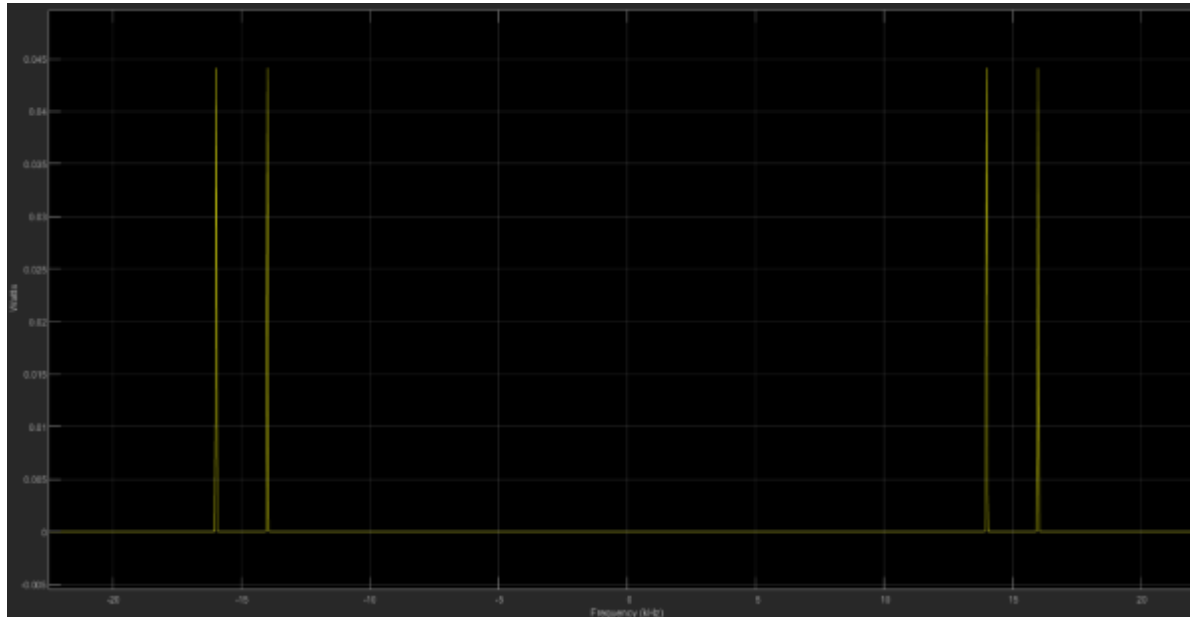


Figure 6: The modulated signal in freq. domain

➤ In time domain:

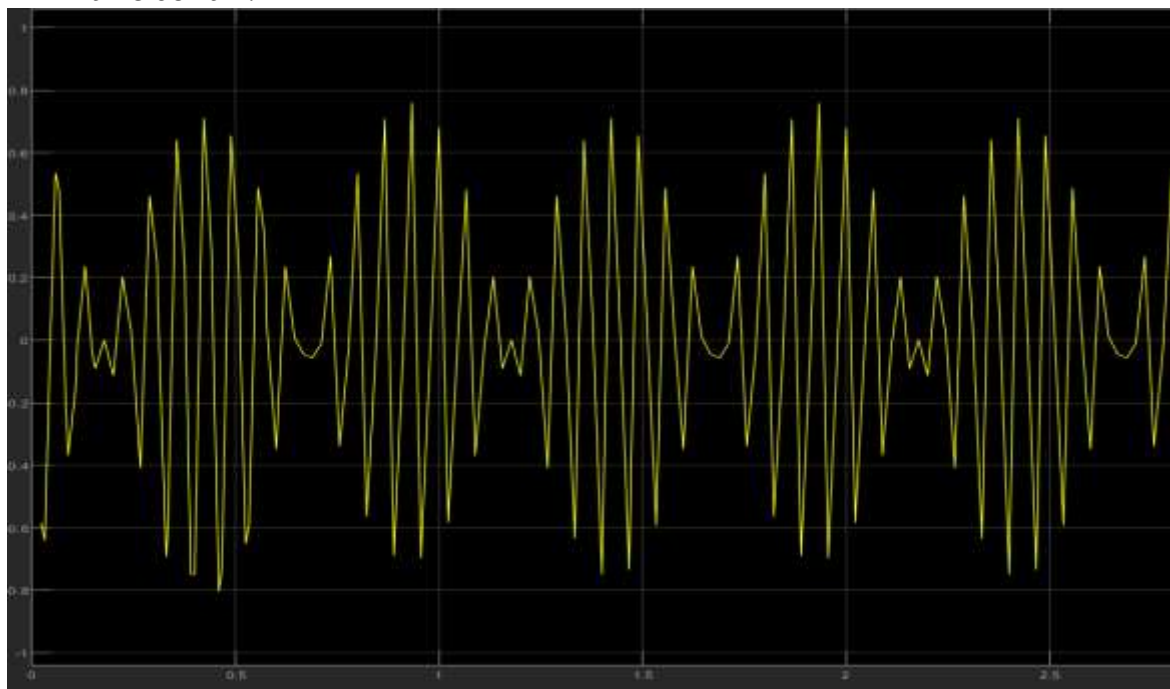


Figure 7: DSBsc modulation in time domain

2. The demodulation :

➤ In frequency domain:

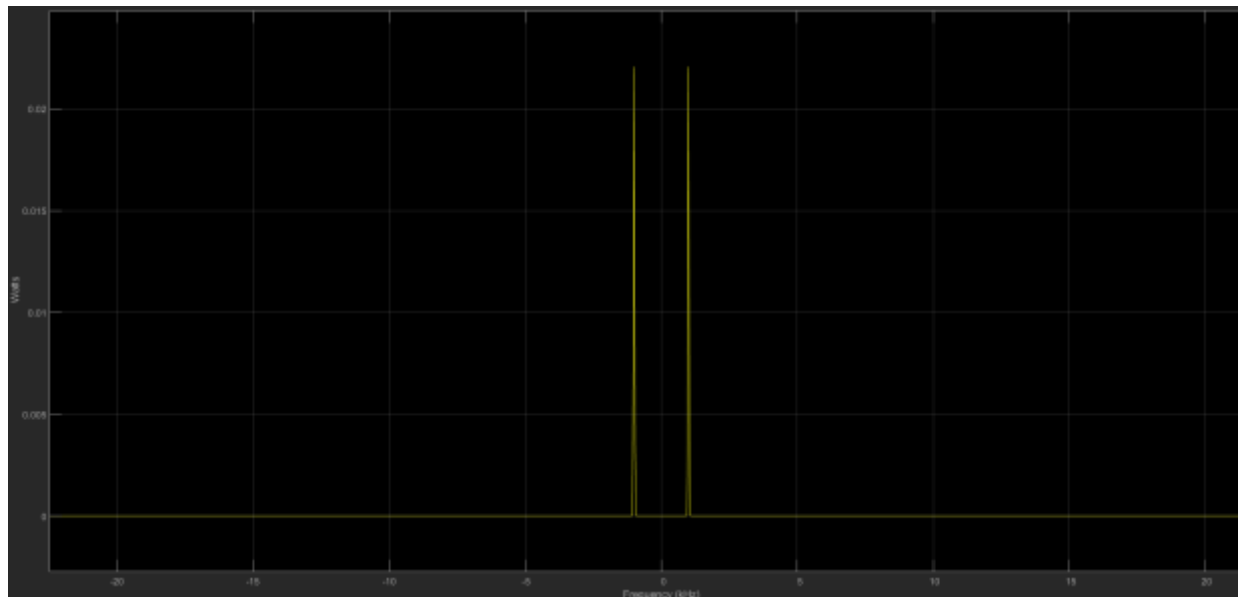


Figure 8: DeModulation output in freq

➤ In time domain:

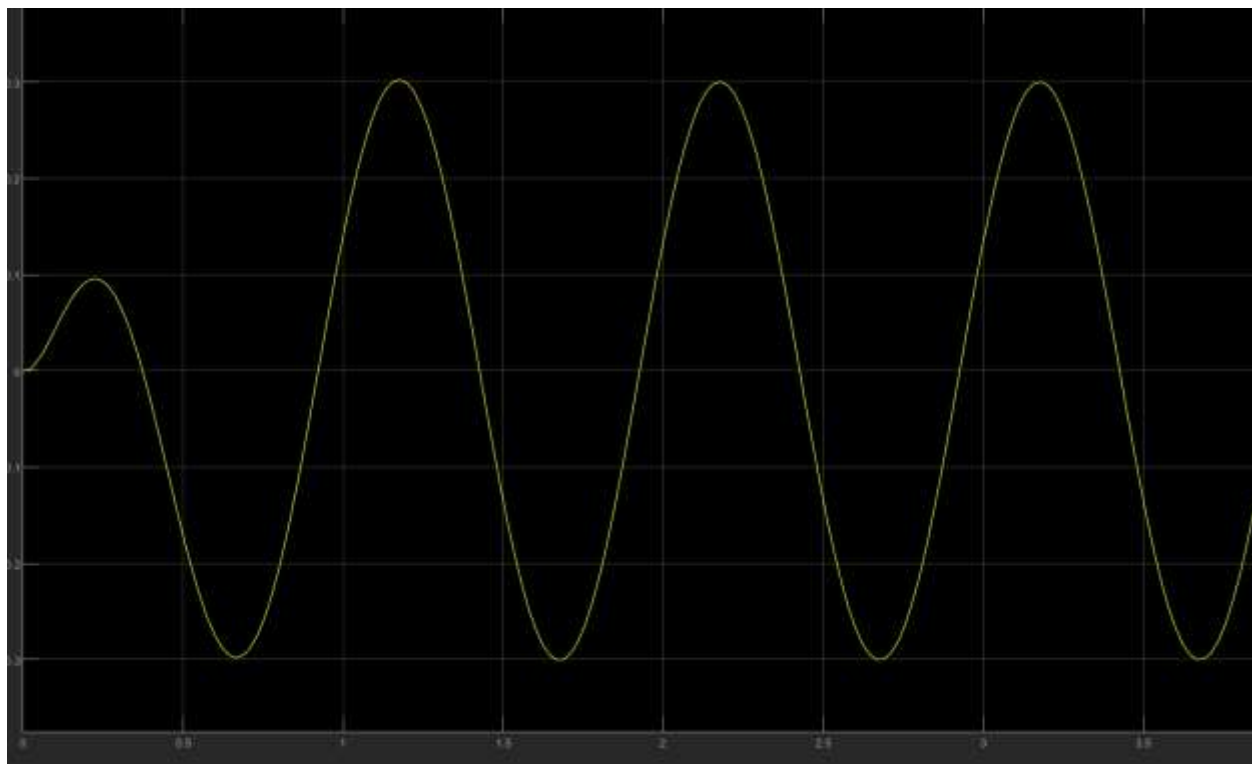


Figure 9: DeModulation output in time

- **SSBsc:**

➤ The block simulation :

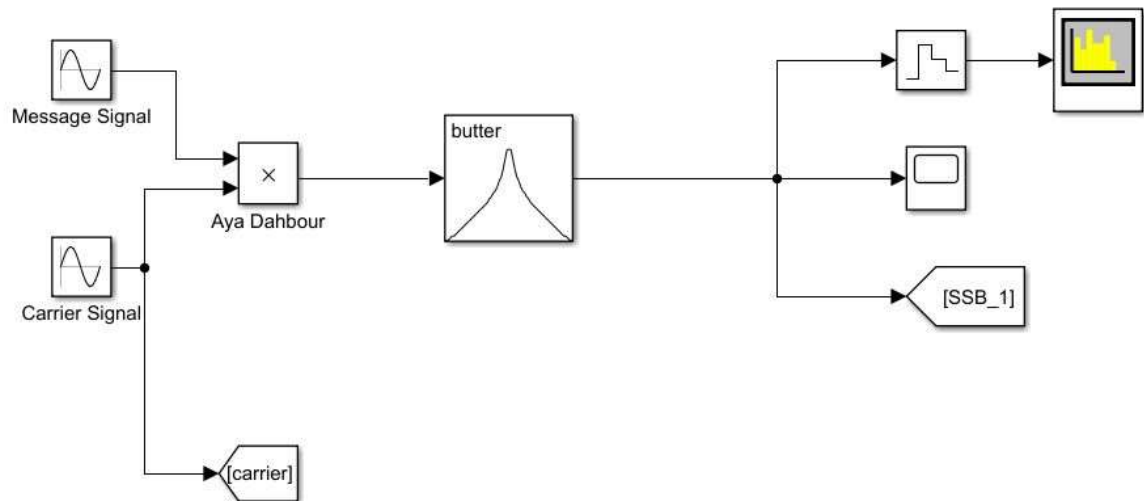


Figure 10:SSBsc modulation block simulation

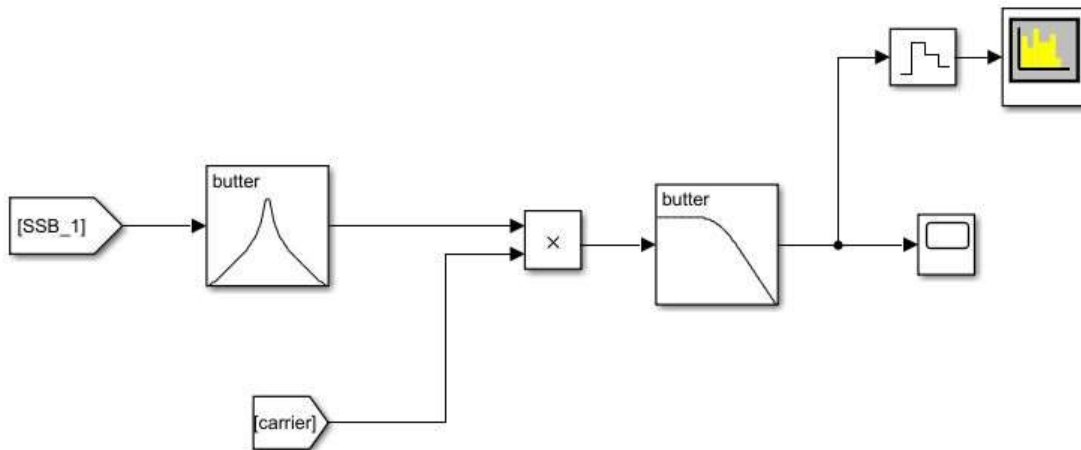


Figure 11:SSBsc demodulation block simulation

After linking the images above and adjusting the signal, filter and measurement settings. I got these results:

1. The modulation :
 - In frequency domain:

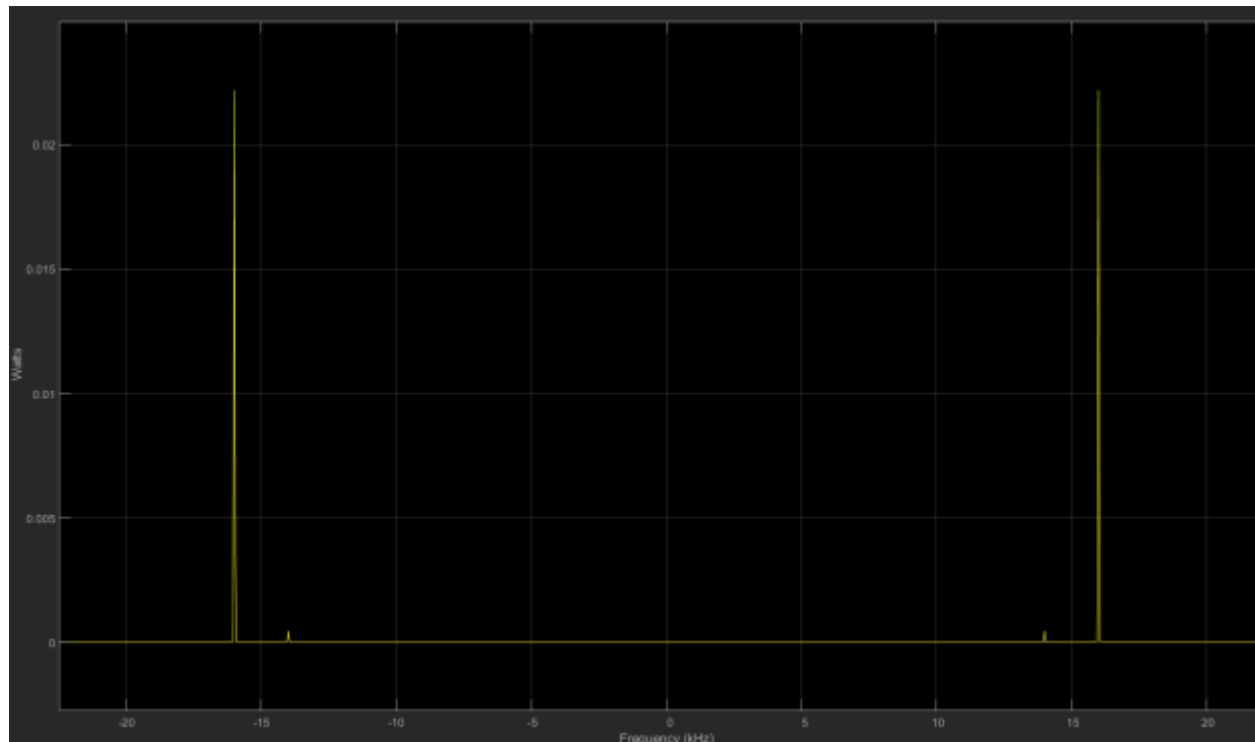


Figure 12:SSBsc modulated in freq

- In time domain :

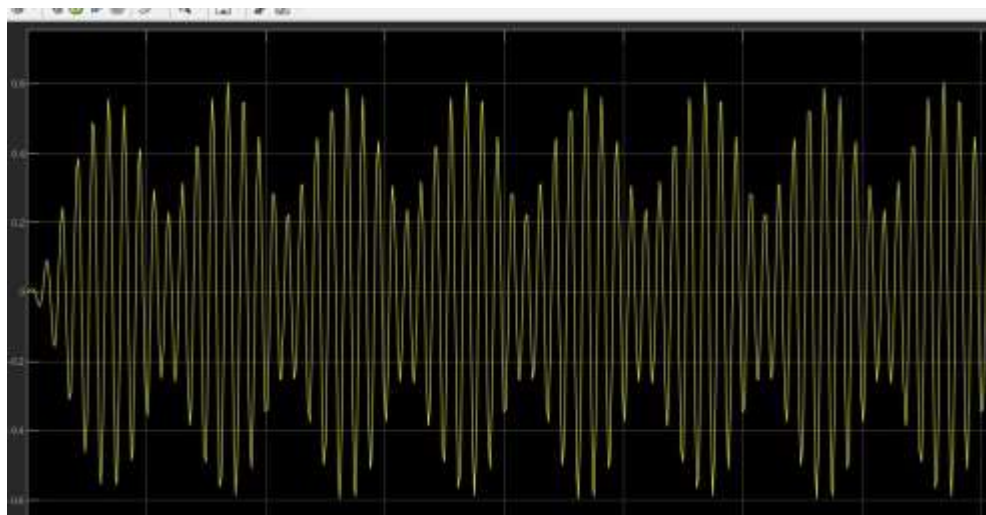


Figure 13: SSB modulated in time domain

2. The demodulation :
- In frequency domain :

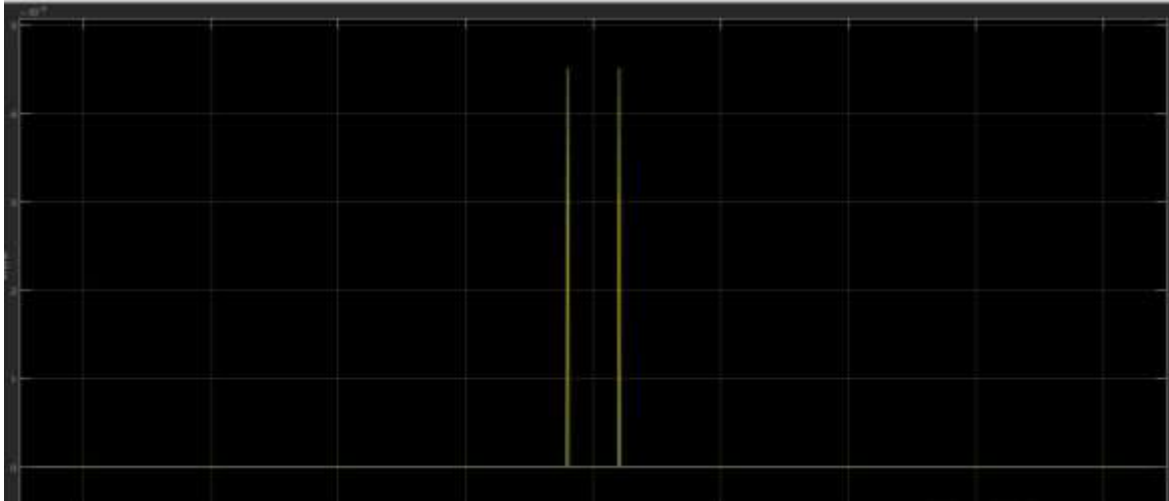


Figure 14: SSBsc demodulation

- In time domain :

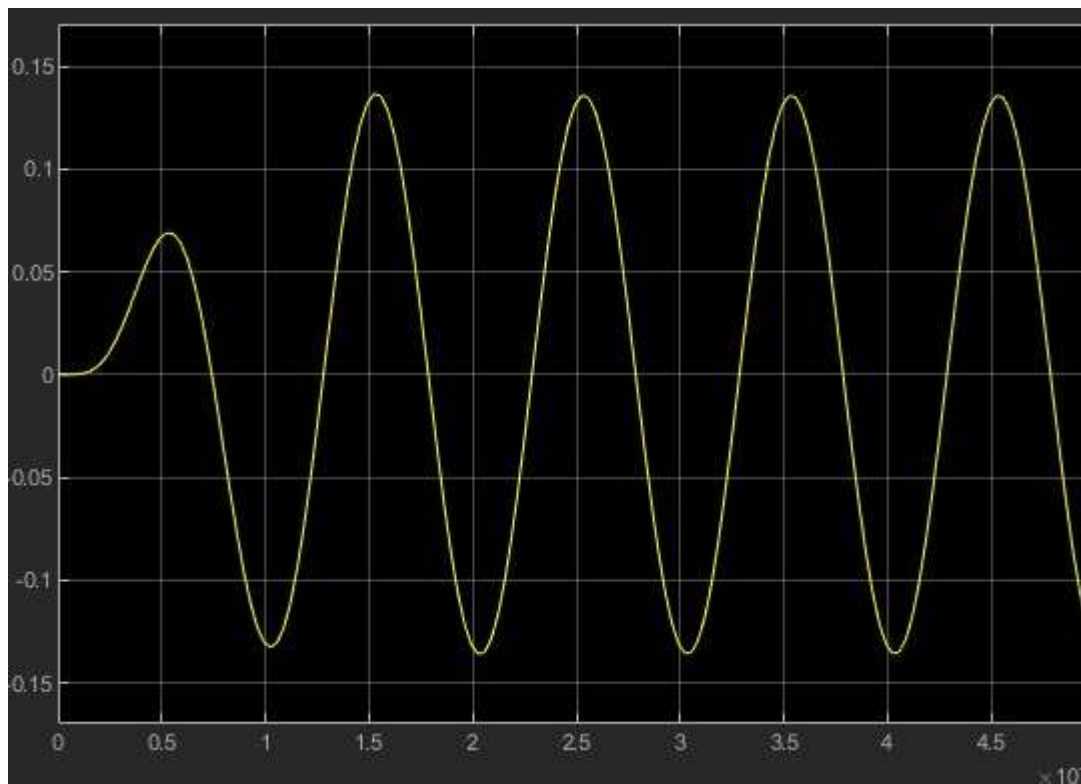


Figure 15:SSBsc demodulated signal in time domain